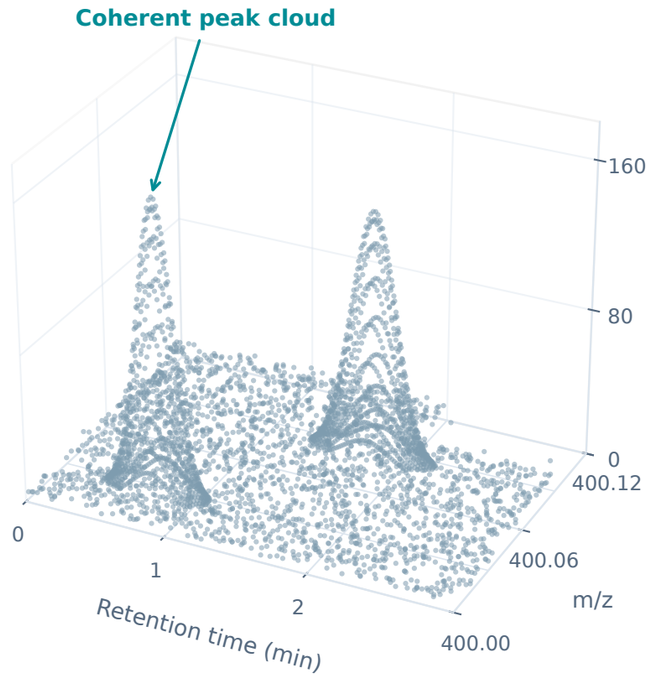
Estimate the local noise floor from isolated centroids

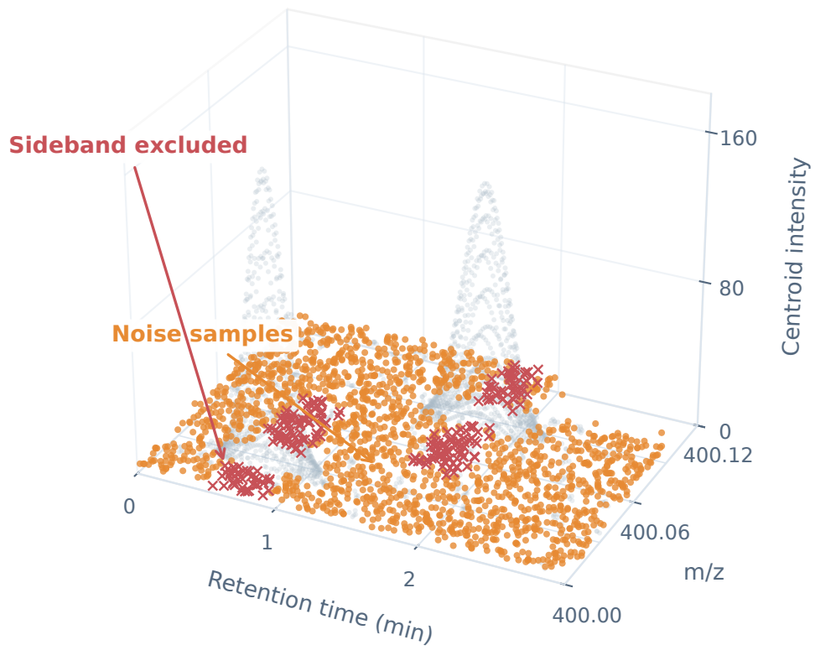
After an initial basin merge, short-lived points supply the noise estimate; coherent peaks do not.

1 Peaks and scattered points



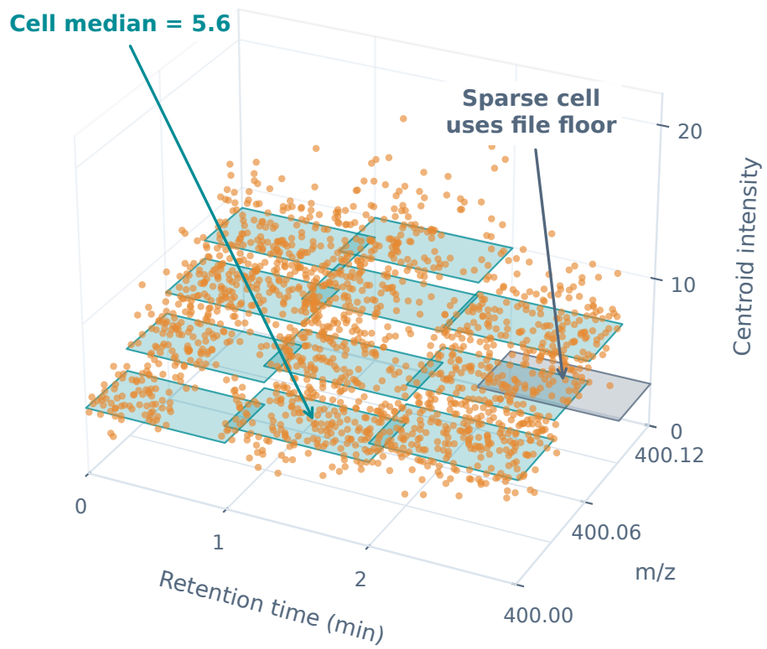
Longer structures hold coherent signal.
Short basins supply candidate noise points.

2 Select noise samples



Use basins spanning at most 2 scans.
Exclude points near much stronger traces.

3 Estimate each cell's floor



Median of eligible raw intensities per cell.
Fewer than 50 points? Use the file's 1st percentile.

Orange = eligible noise samples • Red crosses = excluded sidebands • Panel 3 enlarges the low-intensity range

Select isolated points → exclude nearby sidebands → take a median in each time–m/z cell.

Synthetic data • exact picker noise-floor calculation • four mass bands shown for clarity (default: 16) • 1-minute cells